Notes: Lee (JFE, 2026)

Face-to-face Social Interactions and Local Informational Advantage

Contents

1 Big picture

- **Question.** Do face-to-face (F2F) social interactions still generate a *causal* local informational advantage for mutual fund managers, even in an environment with rich public disclosure and modern remote communication?
- **Setting.** The paper studies U.S. active equity mutual funds and their local versus distant stock holdings around the COVID-19 lockdowns of 2020.
- **Core challenge.** Geographic proximity is correlated with many omitted characteristics—manager skill, risk tolerance, resources, and general social interaction intensity. So a standard "local investors do better on local stocks" fact does not isolate the role of the *F2F channel* itself.
- **Research design.** The paper uses stay-at-home orders, foot-traffic collapses, and local COVID case counts as shocks that curtailed in-person meetings. The empirical idea is to compare local versus distant investments before and after these shocks.
- **Main result.** Local informational advantage weakened when F2F contact was disrupted. After lockdowns, monthly benchmark-adjusted and DGTW-adjusted returns on local portfolios were about 0.2 to 0.4 percentage points lower than those on distant portfolios relative to the pre-lockdown period.
- **Mechanism result 1: trust-building.** The decline was stronger when generalized trust between fund managers and corporate managers was lower. This is consistent with F2F interaction helping transmit trust-sensitive soft information.
- **Mechanism result 2: impression management.** The decline was stronger in *buy* decisions than sell decisions, and stronger for firms whose CEOs had higher pay-for-performance sensitivity. This is consistent with F2F interaction helping the transmission of favorable information.
- **Investment-timing result.** On the same stock, local funds increased portfolio weight by about 0.02 percentage points less than distant funds per unit increase in stock returns after the lockdowns. A one-percentage-point increase in a local position was then associated with about 0.9 to 1.5 percentage points lower abnormal returns over the next one to three months.
- **Why this paper matters.** The paper gives unusually clean causal evidence that F2F interaction is not redundant. It matters for how soft information moves from firms to investors, even in modern financial markets.

2 Data and measurement (key objects)

2.1 Observed mutual fund panel

- **Fund universe.** The sample starts from the CRSP Survivor-Bias-Free Mutual Fund database and Morningstar. It keeps U.S. *active equity* mutual funds operating both before and after the lockdowns.
- **Fund filters.** The paper keeps Morningstar style-box equity funds and excludes non-equity funds, index funds, ETFs, international and regional funds, balanced funds, and sector funds. It also keeps funds with more than 50% in common or preferred stocks and between 20 and 500 holdings.

- **Final fund sample.** These restrictions yield **1,064 funds with distinct portfolios**.
- **Time period.** The core sample runs from **January 2019 to December 2020**, chosen to capture the early pandemic period when in-person interaction was abruptly curtailed.
- **Monthly returns.** The paper computes fund returns, benchmark-adjusted returns, and Daniel–Grinblatt–Titman–Wermers (DGTW) adjusted returns.
- **Fund characteristics.** Total net assets (TNA), flows, and local overweight are measured monthly.

2.2 Observed stock panel and portfolio segmentation

- **Stock data.** Holdings are matched to CRSP and Compustat. The paper keeps U.S. firms with share codes 10 or 11 and nonmissing CRSP and Compustat information.
- **Final firm sample.** The resulting stock sample contains **2,983 firms**.
- **Local versus distant stocks.** A stock is *local* if its headquarters is within **100 miles** of the fund manager's location. Otherwise it is distant.
- **Multiple manager locations.** If a fund has managers in more than one city, a stock is treated as local if it lies within 100 miles of *any* relevant manager location.
- **Portfolio split.** Each fund portfolio is decomposed into a local segment and a distant segment, and the paper studies their value-weighted returns separately.

2.3 Fund manager location data

- **Why manager location matters.** Day-to-day investment decisions are made by managers, not necessarily at the headquarters of the fund family. So using the headquarters address alone can mismeasure the geography of F2F interaction.
- **Location source.** Exact manager locations are hand-collected from the **LexisNexis Public Records Database** and cross-checked with LinkedIn.
- **Coverage.** Residential ZIP codes are retrieved for about **69%** of sample funds. If residence is unavailable, the paper uses regional office ZIP codes identified from LinkedIn.
- **Multiple-location funds.** More than **25%** of sample funds are linked to multiple locations, with an average of **1.7 cities** per fund.
- **Geographic coverage.** Fund managers are located across **41 states** and the District of Columbia; firm headquarters span **48 states** and the District of Columbia.

2.4 Lockdown and mobility measures

- **Measure 1: stay-at-home orders.** The first lockdown proxy is based on state and county stay-at-home orders. A fund or firm is treated as under lockdown once its ZIP code is subject to an order.

- **Measure 2: foot traffic.** The second proxy uses SafeGraph Patterns data on visits to more than 3.6 million commercial points of interest, drawn from about 45 million smartphones. The sample is roughly 10% of the U.S. population.
- **Operational definition.** A ZIP code is classified as under lockdown once foot traffic falls by at least **50%** relative to its 2019 average.
- **Magnitude of mobility collapse.** Average foot-traffic reductions were about −25% in March 2020, −62% in April, and −50% in May.
- **Measure 3: infection risk.** The third proxy is county-level COVID-19 case counts, capturing voluntary avoidance of in-person contact driven by local infection risk.

2.5 Mechanism variables

- **Trust Distance (TD).** The paper infers countries of origin of CEOs, top executives, and fund managers from surnames using NamSor, then matches bilateral trust scores from the Eurobarometer survey. Higher values of TD mean *lower generalized trust*.
- **CEO Delta.** CEO pay-for-performance sensitivity from Core and Guay (2002) proxies for the incentive to emphasize favorable information.
- **Within-family information flow.** The Speed of Information Diffusion (SID) from Cici et al. (2016) measures how quickly stock ideas spread across affiliated funds inside a family.
- **Public information use.** The Reliance on Public Information (RPI) measure from Kacperczyk and Seru (2007) uses analyst recommendation changes to measure how much trading is explained by public information.
- **Informational environment.** The paper studies heterogeneity by firm size, media mentions, Amihud illiquidity, idiosyncratic volatility, analyst forecast dispersion, and S&P 500 membership.
- **Regional social environment.** The paper uses a county-level social index, job density / destination accessibility, and survey-based mask-use behavior to proxy for how much local environments facilitate F2F interaction.

2.6 Summary statistics

- **Pre-lockdown means.** Monthly raw fund return is 1.83%, benchmark-adjusted return is 0.68%, DGTW-adjusted return is 0.57%, average TNA is about $483 million, and average local overweight is 1.44 percentage points.
- **Post-lockdown means.** Monthly raw return rises to 4.84%, benchmark-adjusted return to 1.18%, DGTW-adjusted return to 0.79%, average TNA to $500 million, and local overweight to 1.66 percentage points.
- **Interpretation.** Average post-lockdown market performance was strong, so the paper's question is not whether funds did poorly in absolute terms. It is whether *local relative advantage* weakened.

3 Models and empirical specifications (all equations + interpretation)

3.1 Conceptual framework: what F2F interaction does

The paper does not build a formal structural model. Instead, it starts from two conceptual mechanisms.

- **Trust-building.** F2F interaction creates rich nonverbal cues, social presence, and stronger generalized trust. This matters especially when information is tacit, noncodified, and hard to verify.
- **Impression management.** F2F interaction also makes it easier to deliver favorable information persuasively, because tone, facial expression, posture, and real-time feedback help shape how messages are received.

These mechanisms yield two predictions:

- **Hypothesis 1.** The adverse impact of losing the F2F channel is larger when generalized trust between fund and corporate managers is lower.
- **Hypothesis 2.** The adverse impact of losing the F2F channel is larger for *buy* decisions than for sell decisions.

3.2 Local and distant portfolio returns

For fund i, local or distant portfolio return is

$$R_{i,t}^{L(D)} = \sum_{j=1}^{L_{i,t}(D_{i,t})} w_{i,j,t}^{L(D)} \, r_{j,t}. \tag{1}$$

- **Interpretation.** The portfolio is split into local and distant holdings, and value weights are re-scaled to sum to one within each segment.
- **Abnormal return measures.** The paper uses benchmark-adjusted returns and DGTW-adjusted returns as outcome variables.

3.3 Baseline difference-in-differences specification for portfolio performance

The main portfolio-level DiD regression is

$$R_{i,t}^{L,D} = \beta_0 + \beta_1 \mathrm{Local}_i + \beta_2 \mathrm{Lockdown}_{i,t} + \beta_3 \big(\mathrm{Local}_i \times \mathrm{Lockdown}_{i,t}\big) + \alpha_i + \lambda_t + \varepsilon_{i,t}. \tag{2}$$

- **Dependent variable.** The return on the local or distant portion of fund i's holdings.
- **Treatment.** $\mathrm{Lockdown}_{i,t}$ switches on when the fund's ZIP code enters a lockdown regime.
- **Key coefficient.** β_3 is the change in local relative to distant performance after the lockdown shock.

- **Fixed effects.** Fund fixed effects absorb time-invariant manager heterogeneity; time fixed effects absorb aggregate market conditions.
- **Interpretation.** A negative β_3 means local portfolios lost their relative edge once F2F interaction was curtailed.

3.4 Stock-level investment timing regression

To rule out the idea that local portfolio underperformance is driven only by worse fundamentals of locally held firms, the paper compares *local and distant investors on the same stock at the same time.* The specification is

$$\begin{aligned}\Delta w_{i,j,t} = \beta_0 &+ \beta_1 \text{Local}_{i,j} + \beta_2 \text{Lockdown}_{j,t} + \beta_3 \text{StockReturn}_{j,t} \\ &+ \beta_4 \big(\text{Local}_{i,j} \times \text{Lockdown}_{j,t}\big) \\ &+ \beta_5 \big(\text{Local}_{i,j} \times \text{StockReturn}_{j,t}\big) \\ &+ \beta_6 \big(\text{StockReturn}_{j,t} \times \text{Lockdown}_{j,t}\big) \\ &+ \beta_7 \big(\text{Local}_{i,j} \times \text{Lockdown}_{j,t} \times \text{StockReturn}_{j,t}\big) \\ &+ \delta_{j,t} + \varepsilon_{i,j,t}.\end{aligned} \tag{3}$$

- **Dependent variable.** Change in fund i's portfolio weight in stock j, or benchmark-relative change in weight.
- **Why stock $\times$ time fixed effects matter.** $\delta_{j,t}$ absorbs all time-varying stock-level fundamentals. So identification comes only from whether *local investors react differently from distant investors to the same stock return at the same time.*
- **Timing logic.** If trades are executed at the prior month's end price, then a larger increase in weight during a positive-return month indicates better timing.
- **Key coefficient.** $\beta_7 < 0$ means local investors became worse timers after the lockdowns.

3.5 Subsequent-return regression: does worse timing show up in future returns?

The paper then asks whether post-lockdown local trades are followed by worse abnormal returns:

$$\begin{aligned}\text{DGTWRET}_{j,t+1} = \beta_0 &+ \beta_1 \text{Local}_{i,j} + \beta_2 \text{Lockdown}_{j,t} + \beta_3 \Delta w_{i,j,t} \\ &+ \beta_4 \big(\text{Local}_{i,j} \times \text{Lockdown}_{j,t}\big) \\ &+ \beta_5 \big(\text{Local}_{i,j} \times \Delta w_{i,j,t}\big) \\ &+ \beta_6 \big(\Delta w_{i,j,t} \times \text{Lockdown}_{j,t}\big) \\ &+ \beta_7 \big(\text{Local}_{i,j} \times \text{Lockdown}_{j,t} \times \Delta w_{i,j,t}\big) \\ &+ \alpha_i + \gamma_j + \lambda_t + \varepsilon_{i,j,t}.\end{aligned} \tag{4}$$

- **Dependent variable.** DGTW-adjusted return next month or over the next three months.
- **Key coefficient.** A negative β_7 means an increase in a local position is followed by lower abnormal returns after the lockdowns, relative to distant positions.
- **Interpretation.** This strengthens the timing story: the problem is not only that local managers changed weights differently, but that those post-lockdown local trades were less profitable.

3.6 Main portfolio-performance result

In Table 3, the interaction term Local $\times$ Lockdown is negative and significant across all specifications.

- Using stay-at-home orders, monthly benchmark-adjusted local returns fall by about 0.41 percentage points relative to distant returns.
- Using the same design with DGTW-adjusted returns, the relative decline is about 0.21 percentage points.
- Using the foot-traffic lockdown definition, the relative declines are about 0.38 and 0.17 percentage points.
- Using county COVID cases, the local relative performance effect is also significantly negative.

Interpretation. The local edge disappears precisely when F2F contact is hardest to sustain.

3.7 Main investment-timing result

In Table 4 Panel A, the triple interaction in equation (3) is negative and significant.

- In the baseline specification, local funds increase portfolio weight by about 0.016 percentage points less than distant funds per unit increase in stock returns after lockdowns.
- Using benchmark-relative weight changes, the effect remains similar: about -0.014 to -0.018.
- The result survives fund fixed effects and controls for fund size and flows.

Interpretation. After F2F interaction is disrupted, local investors respond less well to information embedded in return movements.

3.8 Main subsequent-return result

In Table 4 Panel B, the triple interaction in equation (4) is also negative and significant.

- For next-month abnormal returns, a one-percentage-point increase in a local position is associated with roughly 0.94 to 1.00 percentage points lower abnormal return relative to distant positions after lockdowns.
- Over the following three months, the effect is even larger: about 1.31 to 1.51 percentage points.
- The result survives fund-by-stock fixed effects and firm controls.

Interpretation. The underperformance is not just composition or noise. It is consistent with genuinely worse local trade timing.

3.9 Trust-building test

The trust extension augments equation (4) with TD:

$$\text{Local}_{i,j} \times \text{Lockdown}_{j,t} \times \Delta w_{i,j,t} \times \text{TD}_{i,j}.$$

The paper also defines trust distance more generally as

$$TD_{ij} = -\sum_{c_1=1}^{2}\sum_{c_2=1}^{2} P_{i,c_1}P_{j,c_2}BT_{c_1,c_2},$$

where BT_{c_1,c_2} is the bilateral trust score from Eurobarometer.

- **Interpretation.** Higher TD means lower generalized trust.
- **Empirical prediction.** If F2F interaction helps build trust, then the local timing deterioration should be larger when TD is higher.
- **Result.** The four-way interaction is around -0.79 in the CEO-based specification and remains negative in broader executive-based specifications.
- **Economic meaning.** A one-standard-deviation increase in trust distance is associated with about 0.79 percentage points lower next-period abnormal return for local relative to distant investments, per one-percentage-point increase in portfolio weight, after the lockdowns.

3.10 Impression-management test

The paper re-estimates equation (4) separately for active buys and sells, and then interacts the timing effect with CEO delta.

- **Buy versus sell.** The triple interaction is statistically significant for *buy* trades but not sell trades.
- **CEO incentives.** The four-way interaction with CEO delta is negative and significant, indicating that the adverse timing effect is stronger when managers have stronger incentives to communicate favorable news.
- **Interpretation.** This is exactly what the impression-management channel predicts: F2F matters most when positive soft information is useful and strategically communicated.

3.11 Trade activity, local preference, and alternative channels

- **Trading activity.** The lockdown lowered Active Share modestly (about -0.989 in Table 7 Panel A, or roughly a 1.3% decline relative to a pre-lockdown average of 78.72), but turnover, trade size, and initiation activity remain economically active. So the main results are not driven by managers simply stopping trading.
- **Local preference.** Funds with low initial local bias increased local overweight by about 0.8 percentage points, while the highest-local-bias tercile moved down by about 1 percentage point relative to the low-bias group. This means local overweighting converged across funds after the lockdowns.
- **Within-family information flow.** SID does not change significantly after lockdowns, and high-SID fund families do not show meaningfully different timing deterioration. Stocks local to colleagues show a negative effect, but only about half the magnitude of the main local effect.
- **Public information.** RPI falls modestly in the short run and low-RPI funds increase reliance on public information somewhat, but the shift is too small to explain the main findings.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in a naive interpretation

- A simple comparison of local and distant portfolio returns could confound informational advantage with differences in portfolio composition.
- A simple cross-sectional comparison of manager performance could also confound location with time-invariant manager characteristics.
- A simple local-portfolio underperformance result during COVID could reflect worse local fundamentals in locked-down areas rather than impaired information transmission.

4.2 What identifies the paper's main design

- **Exogenous timing shock.** COVID lockdowns and mobility collapses provide sharp, plausibly exogenous reductions in F2F contact.
- **Fixed geographic proximity.** The distance between manager and firm is predetermined, so the shock changes the *feasibility of F2F contact* while holding proximity itself fixed.
- **Difference-in-differences.** The local-versus-distant comparison before and after lockdowns isolates the incremental role of F2F interaction beyond generic pandemic effects.

4.3 What identifies the stock-level timing design

- The stock-level timing regression adds **stock $\times$ time fixed effects**. This is the key move in the paper.
- These fixed effects absorb any contemporaneous stock-specific fundamentals, sectoral shocks, and local economic conditions affecting that stock in that month.
- Therefore, identification comes from comparing *local and distant investors trading the same stock at the same time*.
- This design sharply reduces the concern that local underperformance merely reflects local firms doing poorly during lockdowns.

4.4 Why the mechanism evidence is informative

- The trust-distance interaction is not just another heterogeneity exercise. It tests a very specific mechanism: whether F2F contact helps overcome low generalized trust.
- The buy-versus-sell asymmetry is also disciplined by theory. If F2F mainly helps the flow of favorable soft information, then the effect should show up much more clearly in purchases than in sales.
- The CEO-delta interaction ties the effect to firms whose managers have stronger incentives to accentuate favorable news.

4.5 Why the results are not just about other information channels

- Within-family information diffusion does not break down enough to explain the main result.
- Public-information seeking changes only modestly.
- The effect is stronger where soft information should matter most: in less transparent firms and in regions where social infrastructure or physical density made F2F contact especially valuable before the pandemic.

4.6 Relation to the literature

- The paper speaks to the literature on local informational advantage in mutual funds, but moves beyond reduced-form geography facts by isolating the F2F channel itself.
- It also contributes to the literature on social interactions and finance by distinguishing *face-to-face* interaction from generic social proximity.
- Finally, it links the finance literature on soft information to the psychology and organizational-behavior literature on trust, nonverbal communication, and impression management.

5 Tables and figures

5.1 Figure 1: locations of fund managers and firm headquarters

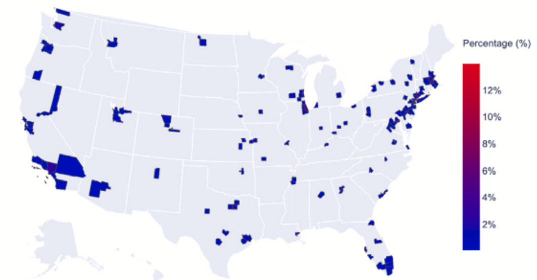
Read:

- The sample has wide national geographic coverage.
- This matters because the paper's design relies on meaningful cross-sectional variation in local versus distant holdings.
- It also shows that manager location is not mechanically the same as fund-family headquarters, which is one of the paper's data contributions.

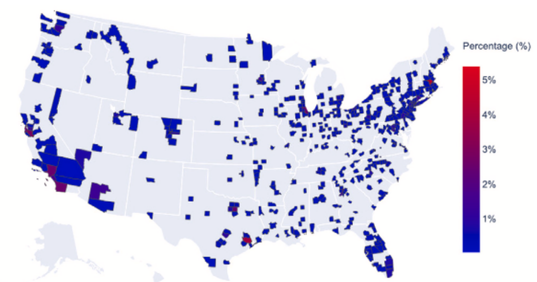
5.2 Figure 2 and Table 2: lockdown rollout and mobility collapse

Read:

- Panel A shows that lockdowns rolled out rapidly across states and counties in March and April 2020.
- Panel B shows a sharp collapse in foot traffic exactly when lockdowns begin, followed by only partial recovery.
- Table 2 quantifies the collapse: roughly -25% in March, -62% in April, and -50% in May.
- This is the empirical foundation for interpreting 2020 as a period of sharply reduced F2F interaction.



(a) Panel A. Fund managers



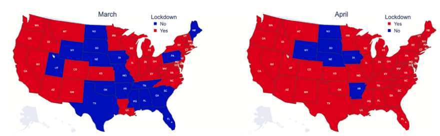
(b) Panel B. Firm headquarters

Fig. 1. Locations of sample fund managers and firm headquarters.
Fig. 1 presents the county-level geographical distribution of sample mutual fund managers in Panel A and sample firm headquarters in Panel B.

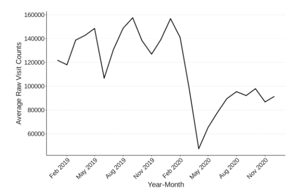
Table 2
Change in foot traffic.

	Mean	Median	St. Dev.	Pctl(5)	Pctl(25)	Pctl(75)	Pctl(95)
Mar 2020	−25.4	−25.02	14.31	−48.87	−33.82	−17.39	−4.58
Apr 2020	−62.43	−63.58	15.42	−87.79	−72.77	−53.27	−37.39
May 2020	−50.29	−50.88	19.95	−84.74	−63.59	−37.79	−16.69
Jun 2020	−40.53	−41.05	22.97	−79.78	−54.94	−26.24	−4.53
Jul 2020	−32.21	−31.37	22.66	−72.81	−45.11	−18.1	2.67
Aug 2020	−27.96	−26.96	22.49	−68.82	−40.98	−13.28	6.55
Sep 2020	−30.22	−29.3	20.67	−67.75	−41.84	−17.17	1.97
Oct 2020	−26.86	−25.71	21.37	−66.04	−38.91	−13.17	6.08
Nov 2020	−36.86	−36.46	19.56	−72.31	−47.96	−24.79	−6.65
Dec 2020	−34.51	−33.66	22.26	−75.13	−48.29	−20.02	1.25

Table 2 reports the percentage change in SafeGraph foot traffic, relative to the 2019 average, for the ZIP codes of sample funds and firms.



(a) Panel A. Stay-at-home orders



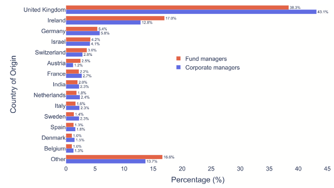
(b) Panel B. Aggregated foot traffic

Fig. 2. Stay-at-home orders and foot traffic activity.
Fig. 2 Panel A shows the adoption of stay-at-home orders across U.S. states for March (left) and April (right) 2020. States with orders in effect are shown in red, and those without are shown in blue. Panel B plots foot traffic activity in the locations of sample firms and funds during the sample period, showing the average monthly total foot traffic across all sample ZIP codes, obtained from the SafeGraph Patterns database.

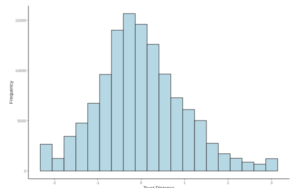
Table 1
Sample fund returns and characteristics.

Panel A. Pre-lockdown	Mean	Median	St. Dev.	Pctl(5)	Pctl(25)	Pctl(75)	Pctl(95)
Return	1.83	2.56	5.43	−8.38	−1.35	5.16	10.01
Benchmark-adjusted return	0.68	0.56	1.62	−1.79	−0.26	1.5	3.59
DGTW-adjusted return	0.57	0.42	1.54	−1.74	−0.36	1.34	3.4
TNA	483.28	91.7	862.74	0	17.27	440.42	3158.26
Flow	0.06	−0.02	0.67	−0.12	−0.06	0.05	0.22
Local overweight	1.44	1.03	5.89	−8.17	−1.57	4.22	11.99
Panel B. Post-lockdown							
Return	4.84	5.12	7.39	−11.3	0.46	10	16.12
Benchmark-adjusted return	1.18	0.94	2.13	−2.16	−0.13	2.32	5.66
DGTW-adjusted return	0.79	0.6	1.93	−2.21	−0.43	1.85	4.63
TNA	500	97.23	875.33	0.13	20.13	478.88	3158.26
Flow	0.05	−0.02	0.56	−0.14	−0.07	0.07	0.28
Local overweight	1.66	1.18	6.16	−8.66	−1.52	4.59	13.18

Table 1 summarizes the characteristics of sample funds reported separately for the pre-lockdown period (from January 2019 through the month before lockdown orders were implemented in the fund's location) in Panel A, and for the post-lockdown period (from the implementation month through December 2020) in Panel B. Returns are calculated monthly and reported in percentage points. TNA refers to the fund's total net assets in millions. $Flow$ denotes monthly fund inflows in decimals. *Local overweight* is a measure from Coval and Moskowitz (2001) in percentage points.



(a) Panel A. Managers' countries of origin



(b) Panel B. Trust Distance (TD)

Fig. 3. Managers' countries of origin and trust distance.
Fig. 3 Panel A shows the distribution of countries of origin for corporate managers (CEOs and the five highest-paid executives) at sample firms, as well as for sample fund managers. Panel B presents the distribution of TD, a standardized measure of trust distance between fund-firm pairs constructed based on Eurobarometer survey data.

5.3 Table 1: sample fund returns and characteristics

Read:

- The typical fund is fairly large, with average TNA around $483 million pre-lockdown and $500 million post-lockdown.
- Local overweight is positive on average both before and after the shock, confirming persistent local tilt.
- The distribution of local overweight is wide, with a right-skewed tail, which motivates the later convergence exercise by local-bias terciles.

5.4 Figure 3: countries of origin and trust distance

Read:

- The figure shows substantial variation in managers' inferred countries of origin and in trust distance across fund-firm pairs.
- This gives meaningful cross-sectional variation for the trust-building mechanism test.
- The distribution is centered near zero after standardization but with a nontrivial spread, so trust heterogeneity is empirically important.

5.5 Table 3 and Figure 4: portfolio-level impact of lockdowns

Read:

- Table 3 shows that local portfolios underperform distant portfolios after lockdowns by about 0.2 to 0.4 percentage points per month in abnormal-return terms.
- The result survives alternative lockdown definitions based on stay-at-home orders, foot traffic, and COVID case counts.

Table 3
Impact of lockdowns on local portfolio performance.

	Dependent variable:					
	Benchmark-adjusted return			DGTW-adjusted return		
	(1)	(2)	(3)	(4)	(5)	(6)
Local	−0.275***	−0.297***	−0.387***	−0.165***	−0.182***	−0.208***
	(0.036)	(0.037)	(0.038)	(0.033)	(0.035)	(0.033)
Lockdown	0.622*			0.628**		
	(0.318)			(0.261)		
Local × Lockdown	−0.410***			−0.205***		
	(0.058)			(0.055)		
Footprint Drop		0.599***			0.847***	
		(0.185)			(0.164)	
Local × Footprint Drop		−0.375***			−0.172***	
		(0.065)			(0.059)	
Covid Cases			−0.004			0.006
			(0.009)			(0.008)
Local × Covid Cases			−0.024***			−0.018***
			(0.007)			(0.007)
Fund FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	47,750	47,750	47,750	47,750	47,750	47,750
R^2	0.082	0.082	0.081	0.086	0.087	0.086

Table 3 presents regression results from Eq. (2). The dependent variables are the benchmark-adjusted returns for local and distant portfolios in Columns (1)–(3) and DGTW-adjusted returns in Columns (4)–(6) in percentage points. *Local* is an indicator equal to one for the local portfolio, defined using a 100-mile threshold. *Lockdown* is an indicator equal to one after the fund's ZIP code becomes subject to county-level stay-at-home orders. *Footprint Drop* is an indicator equal to one after the fund's ZIP code experiences a foot traffic decline of at least 50% relative to the 2019 average. *COVID Cases* denotes county-level confirmed COVID-19 cases per 10 million population. Standard errors are clustered at the fund level. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

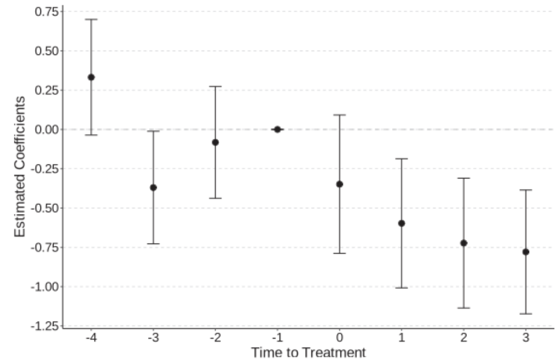


Fig. 4. Impact of lockdowns on local portfolio performance.
Fig. 4 presents the parallel trends check result for Eq. (2). It plots the point estimates of γ_s and the 90 percent confidence intervals, using standard errors clustered at the fund level, for the following regression equation:

$$R_{i,t}^{L,D} = \beta_0 + \beta_1 Local_i + \sum_{s=t-4}^{t+3} \left(\beta_s Event_{i,s} + \gamma_s Local_i \times Event_{i,s}\right) + \alpha_i + \lambda_t + \epsilon_{i,t}$$

where $R_{i,t}^{L,D}$ is the benchmark-adjusted return on the local or distant portion of the fund's holdings; $Local_i$ is an indicator variable equal to one for the local portfolio, defined using a 100-mile threshold; and $Event_{i,s}$ is a time indicator relative to the lockdown month in which foot traffic drops by more than 50% relative to its 2019 average in fund i's ZIP code. The coefficients are compared to those of the month prior to the lockdowns.

- Figure 4 shows no obvious differential pre-trend, which is an important credibility check for the DiD design.
- The portfolio-level result is economically meaningful: with an average local portfolio of about \$277 million, a 0.4-point monthly return loss implies about \$13.3 million annualized per fund.

5.6 Table 4 and Figure 5: investment timing

Read:

- Table 4 Panel A shows that local investors react less effectively than distant investors to the same stock return after lockdowns.
- Table 4 Panel B shows that these post-lockdown local trades subsequently earn lower abnormal returns, especially over the next three months.
- Figure 5 reports the event-study style checks and supports parallel trends before treatment.
- This is the paper's strongest evidence that the mechanism is about impaired information processing and timing, not simply worse local fundamentals.

5.7 Table 5: trust-building mechanism

Read:

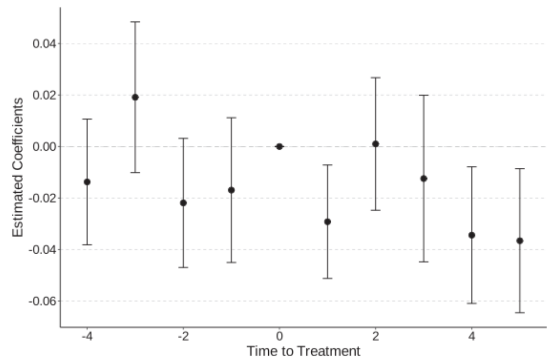
- The interaction with Trust Distance is negative and significant: low-trust relationships are hit harder when F2F contact disappears.
- The effect is strongest in newer relationships (shorter holding periods), which is exactly where generalized trust should matter most.

Table 4
Impact of lockdowns on local investment timing.

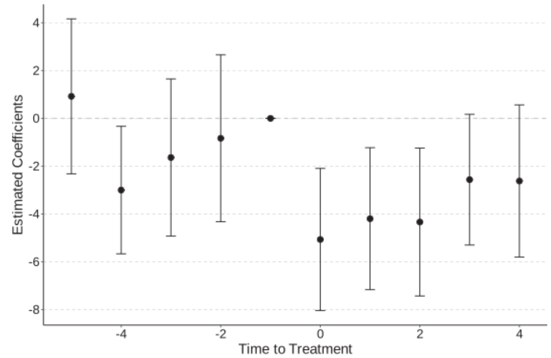
Panel A. Local versus distant investors' portfolio weight adjustment						
	Dependent variable:					
	Δw			ΔExcess w		
	(1)	(2)	(3)	(4)	(5)	(6)
Local	−0.00005	−0.00001	0.00003	−0.0002	−0.0001	−0.0001
	(0.0004)	(0.0005)	(0.0005)	(0.0005)	(0.0005)	(0.0005)
TNA			0.000			0.000
			(0.000)			(0.000)
Flow			−0.0004			−0.0001
			(0.0003)			(0.0005)
Local × Footprint Drop	0.002*	0.002*	0.002**	0.002	0.002	0.002*
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Local × Stock Return	−0.034***	−0.034***	−0.038***	−0.033***	−0.033***	−0.036***
	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)
Local × Footprint Drop × Stock Return	−0.016**	−0.016**	−0.019**	−0.014*	−0.014*	−0.018**
	(0.007)	(0.007)	(0.008)	(0.008)	(0.008)	(0.008)
Stock × Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Fund FE		Yes	Yes		Yes	Yes
Observations	4,078,047	4,078,047	3,857,850	4,078,047	4,078,047	3,857,850
R^2	0.115	0.115	0.115	0.083	0.084	0.084

Panel B. Subsequent-period return of local versus distant investments						
	Dependent variable:					
	$DGTW_{RET,j,t+1}$			$DGTW_{RET,j,t+3}$		
	(1)	(2)	(3)	(4)	(5)	(6)
Local	0.072			0.317*		
	(0.062)			(0.182)		
Lockdown	−1.142**	−1.328**	−1.094**	−0.390	−0.647	−0.280
	(0.529)	(0.555)	(0.531)	(0.872)	(0.864)	(0.825)
Δw	−1.049***	−1.392***	−0.096	−1.951***	−2.795***	−0.340*
	(0.131)	(0.144)	(0.121)	(0.223)	(0.245)	(0.197)
Local × Lockdown	−0.141	−0.211	−0.222	−0.695*	−0.864**	−0.870**
	(0.134)	(0.144)	(0.160)	(0.387)	(0.404)	(0.434)
Local × Δw	0.409*	0.337	0.357	0.679*	0.461	0.436
	(0.223)	(0.237)	(0.233)	(0.347)	(0.366)	(0.343)
Lockdown × Δw	0.234	0.304	0.461**	0.838***	1.195***	1.474***
	(0.196)	(0.216)	(0.207)	(0.313)	(0.342)	(0.348)
Local × Lockdown × Δw	−0.940***	−0.995***	−0.939***	−1.510***	−1.416***	−1.313**
	(0.306)	(0.336)	(0.320)	(0.461)	(0.514)	(0.510)
Fund FE	Yes			Yes		
Fund × Stock FE		Yes	Yes		Yes	Yes
Stock FE	Yes			Yes		
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm Controls			Yes			Yes
Observations	2,333,676	2,333,676	2,320,003	2,325,119	2,325,119	2,314,837
R^2	0.054	0.088	0.103	0.148	0.223	0.241

Table 4 Panel A reports the results from estimating Equation (3). The dependent variable is the change in the fund's portfolio weight in Columns (1)–(3) and the benchmark-relative change in portfolio weight in Columns (4)–(6). *Local* is an indicator equal to one if the distance between the fund and the firm is less than 100 miles. *Footprint Drop* indicates the post-lockdown period for the firm's headquarters based on foot traffic. *Stock Return* is the stock's return during the month in decimals. TNA and $Flow$ are the fund's monthly total net assets and net flow. The sample includes all fund-stock pairs with non-zero holdings at some point, assigning a weight of zero in months when the fund holds no position. Panel B reports the results from estimating Equation (4). The dependent variable is the DGTW-adjusted return of stock j in the next month in Columns (1)–(3) and over the subsequent three months in Columns (4)–(6). Firm-level control variables include total assets, market capitalization, book-to-market ratio, return on assets, and contemporaneous DGTW-adjusted returns. The sample includes all fund-stock pairs with non-zero holdings. Standard errors are clustered at the fund and stock levels. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.



(a) Panel A. Local versus distant investors' portfolio weight adjustment



(b) Panel B. Subsequent three-month return of local versus distant investments

Fig. 5. Impact of lockdowns on local investment timing.
Fig. 5 Panel A presents the parallel trends check result for Eq. (3), and Panel B does so for Eq. (4). Both panels plot the point estimates and 90% confidence intervals for the three-way interaction term from an event-time specification, following the approach in Fig. 4. In Panel A, coefficients are compared to the lockdown month, given the assumption that trades are executed at the end-of-month price of the preceding month. In Panel B, coefficients are compared to the month before the lockdowns.

Table 5
Heterogeneity by trust between corporate and fund managers.

Panel A. Local investment timing and trust distance						
	Dependent variable:					
	$DGTW_{RET,j,t+1}$					
	CEO			Top five executives		
	(1)	(2)	(3)	(4)	(5)	(6)
Local	0.030			0.033		
	(0.073)			(0.065)		
Lockdown	-1.300^{**}	-1.478^{**}	-1.211^{**}	-1.218^{**}	-1.391^{**}	-1.119^{*}
	(0.606)	(0.635)	(0.609)	(0.581)	(0.608)	(0.584)
Δw	-0.931^{***}	-1.272^{***}	-0.213	-0.917^{***}	-1.243^{***}	-0.151
	(0.149)	(0.162)	(0.135)	(0.142)	(0.155)	(0.126)
TD	0.028			0.026		
	(0.033)			(0.020)		
Local × Lockdown	-0.048	-0.085	-0.029	-0.030	-0.067	-0.027
	(0.165)	(0.174)	(0.187)	(0.148)	(0.156)	(0.169)
Local × Δw	0.402	0.300	0.252	0.286	0.220	0.184
	(0.246)	(0.260)	(0.258)	(0.235)	(0.248)	(0.242)
Lockdown × Δw	0.389^{*}	0.489^{*}	0.641^{**}	0.271	0.369	0.505^{**}
	(0.235)	(0.261)	(0.251)	(0.213)	(0.237)	(0.228)
Local × TD	-0.047			-0.016		
	(0.056)			(0.034)		
Lockdown × TD	-0.082	-0.104	-0.109	-0.048	-0.054	-0.050
	(0.082)	(0.089)	(0.095)	(0.047)	(0.053)	(0.059)
Δw × TD	-0.049	-0.099	-0.082	-0.018	-0.029	-0.013
	(0.089)	(0.092)	(0.091)	(0.070)	(0.078)	(0.073)
Local × Lockdown × Δw	-0.742^{**}	-0.692^{*}	-0.666^{*}	-0.603^{*}	-0.606^{*}	-0.596^{*}
	(0.338)	(0.367)	(0.358)	(0.319)	(0.347)	(0.338)
Local × Lockdown × TD	0.041	0.021	0.064	0.003	0.013	0.039
	(0.127)	(0.137)	(0.146)	(0.076)	(0.082)	(0.088)
Local × Δw × TD	0.245	0.166	0.148	0.321^{**}	0.284^{*}	0.256
	(0.185)	(0.187)	(0.193)	(0.153)	(0.163)	(0.163)
Lockdown × Δw × TD	0.151	0.181	0.174	0.077	0.086	0.062
	(0.133)	(0.146)	(0.143)	(0.103)	(0.115)	(0.112)
Local × Lockdown × Δw × TD	-0.791^{***}	-0.787^{***}	-0.772^{***}	-0.432^{*}	-0.439^{*}	-0.400
	(0.260)	(0.276)	(0.274)	(0.239)	(0.258)	(0.258)
Fund FE	Yes			Yes		
Fund × Stock FE		Yes	Yes		Yes	Yes
Stock FE	Yes			Yes		
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm Controls			Yes			Yes
Observations	1,477,414	1,477,414	1,472,196	1,730,581	1,730,581	1,725,014
R^2	0.049	0.077	0.089	0.049	0.076	0.088

Table 5 (*continued*).

Panel B. Local investment timing and trust distance for different holding periods						
	Dependent variable:					
	$DGTW_{RET,j,t+1}$					
	1 year		3 years		5 years	
	$<$	$>$	$<$	$>$	$<$	$>$
	(1)	(2)	(3)	(4)	(5)	(6)
Local × Lockdown × Δw × TD	-0.809^{*}	-0.439	-0.774^{***}	-0.271	-0.699^{**}	-0.239
	(0.423)	(0.315)	(0.286)	(0.343)	(0.275)	(0.393)
Fund FE	Yes	Yes	Yes	Yes	Yes	Yes
Stock FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	295,954	1,178,065	673,842	800,177	937,932	536,087
R^2	0.058	0.050	0.052	0.049	0.050	0.053

Panel C. Local investment timing and trust distance controlling for shared backgrounds						
	Dependent variable:					
	$DGTW_{RET,j,t+1}$					
	(1)	(2)	(3)	(4)	(5)	(6)
Local × Lockdown × Δw × TD	-0.802^{***}	-0.699^{**}	-0.791^{***}	-0.799^{***}	-0.689^{**}	-0.783^{***}
	(0.267)	(0.286)	(0.260)	(0.265)	(0.286)	(0.259)
Fund FE	Yes	Yes	Yes	Yes	Yes	Yes
Stock FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Cultural Distance	Yes	Yes				
I(Common Education)			Yes	Yes		
I(Common Gender)					Yes	Yes
I(Common Race)					Yes	Yes
Firm Controls		Yes		Yes		Yes
Observations	1,428,156	1,423,094	1,118,743	1,114,287	1,478,042	1,472,808
R^2	0.049	0.057	0.046	0.053	0.049	0.057

Table 5 Panel A reports regression results that extend Equation (4) by introducing *Trust Distance* (TD), a trust measure constructed for each firm-fund pair based on Eurobarometer survey results. In Columns (1)–(3), TD is measured using the inferred countries of origin of the CEO and fund managers. In Columns (4)–(6), the measure is based on the top five highest-paid executives at each firm. Panel B presents estimates of the final interaction term from Panel A, separately for subsamples defined by holding periods of 1, 3, and 5 years. Panel C reports results from Panel A with additional controls for shared backgrounds. *Cultural Distance* is from Berry et al. (2010). $I(Common\ Education)$, $I(Common\ Gender)$, and $I(Common\ Race)$ are indicator variables equal to one if at least one firm-fund manager pair shares the same undergraduate or MBA institution, gender, or race, respectively. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

- The result survives controls for cultural distance, common education, gender, and race, so it is not just generic homophily.

Table 5
Heterogeneity by trust between corporate and fund managers.

Panel A. Local investment timing and trust distance						
	Dependent variable:					
	$DGTW_{RET,j,t+1}$					
	CEO			Top five executives		
	(1)	(2)	(3)	(4)	(5)	(6)
Local	0.030			0.033		
	(0.073)			(0.065)		
Lockdown	-1.300^{**}	-1.478^{**}	-1.211^{**}	-1.218^{**}	-1.391^{**}	-1.119^{*}
	(0.606)	(0.635)	(0.609)	(0.581)	(0.608)	(0.584)
Δw	-0.931^{***}	-1.272^{***}	-0.213	-0.917^{***}	-1.243^{***}	-0.151
	(0.149)	(0.162)	(0.135)	(0.142)	(0.155)	(0.126)
TD	0.028			0.026		
	(0.033)			(0.020)		
Local × Lockdown	-0.048	-0.085	-0.029	-0.030	-0.067	-0.027
	(0.165)	(0.174)	(0.187)	(0.148)	(0.156)	(0.169)
Local × Δw	0.402	0.300	0.252	0.286	0.220	0.184
	(0.246)	(0.260)	(0.258)	(0.235)	(0.248)	(0.242)
Lockdown × Δw	0.389^{*}	0.489^{*}	0.641^{**}	0.271	0.369	0.505^{**}
	(0.235)	(0.261)	(0.251)	(0.213)	(0.237)	(0.228)
Local × TD	-0.047			-0.016		
	(0.056)			(0.034)		
Lockdown × TD	-0.082	-0.104	-0.109	-0.048	-0.054	-0.050
	(0.082)	(0.089)	(0.095)	(0.047)	(0.053)	(0.059)
Δw × TD	-0.049	-0.099	-0.082	-0.018	-0.029	-0.013
	(0.089)	(0.092)	(0.091)	(0.070)	(0.078)	(0.073)
Local × Lockdown × Δw	-0.742^{**}	-0.692^{*}	-0.666^{*}	-0.603^{*}	-0.606^{*}	-0.596^{*}
	(0.338)	(0.367)	(0.358)	(0.319)	(0.347)	(0.338)
Local × Lockdown × TD	0.041	0.021	0.064	0.003	0.013	0.039
	(0.127)	(0.137)	(0.146)	(0.076)	(0.082)	(0.088)
Local × Δw × TD	0.245	0.166	0.148	0.321^{**}	0.284^{*}	0.256
	(0.185)	(0.187)	(0.193)	(0.153)	(0.163)	(0.163)
Lockdown × Δw × TD	0.151	0.181	0.174	0.077	0.086	0.062
	(0.133)	(0.146)	(0.143)	(0.103)	(0.115)	(0.112)
Local × Lockdown × Δw × TD	-0.791^{***}	-0.787^{***}	-0.772^{***}	-0.432^{*}	-0.439^{*}	-0.400
	(0.260)	(0.276)	(0.274)	(0.239)	(0.258)	(0.258)
Fund FE	Yes			Yes		
Fund × Stock FE		Yes	Yes		Yes	Yes
Stock FE	Yes			Yes		
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm Controls			Yes			Yes
Observations	1,477,414	1,477,414	1,472,196	1,730,581	1,730,581	1,725,014
R^2	0.049	0.077	0.089	0.049	0.076	0.088

Table 5 (*continued*).

Panel B. Local investment timing and trust distance for different holding periods						
	Dependent variable:					
	$DGTW_{RET,j,t+1}$					
	1 year		3 years		5 years	
	$<$	$>$	$<$	$>$	$<$	$>$
	(1)	(2)	(3)	(4)	(5)	(6)
Local × Lockdown × Δw × TD	-0.809^{*}	-0.439	-0.774^{***}	-0.271	-0.699^{**}	-0.239
	(0.423)	(0.315)	(0.286)	(0.343)	(0.275)	(0.393)
Fund FE	Yes	Yes	Yes	Yes	Yes	Yes
Stock FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	295,954	1,178,065	673,842	800,177	937,932	536,087
R^2	0.058	0.050	0.052	0.049	0.050	0.053

Panel C. Local investment timing and trust distance controlling for shared backgrounds						
	Dependent variable:					
	$DGTW_{RET,j,t+1}$					
	(1)	(2)	(3)	(4)	(5)	(6)
Local × Lockdown × Δw × TD	-0.802^{***}	-0.699^{**}	-0.791^{***}	-0.799^{***}	-0.689^{**}	-0.783^{***}
	(0.267)	(0.286)	(0.260)	(0.265)	(0.286)	(0.259)
Fund FE	Yes	Yes	Yes	Yes	Yes	Yes
Stock FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Cultural Distance	Yes	Yes				
I(Common Education)			Yes	Yes		
I(Common Gender)					Yes	Yes
I(Common Race)					Yes	Yes
Firm Controls		Yes		Yes		Yes
Observations	1,428,156	1,423,094	1,118,743	1,114,287	1,478,042	1,472,808
R^2	0.049	0.057	0.046	0.053	0.049	0.057

Table 5 Panel A reports regression results that extend Equation (4) by introducing *Trust Distance* (TD), a trust measure constructed for each firm-fund pair based on Eurobarometer survey results. In Columns (1)–(3), TD is measured using the inferred countries of origin of the CEO and fund managers. In Columns (4)–(6), the measure is based on the top five highest-paid executives at each firm. Panel B presents estimates of the final interaction term from Panel A, separately for subsamples defined by holding periods of 1, 3, and 5 years. Panel C reports results from Panel A with additional controls for shared backgrounds. *Cultural Distance* is from Berry et al. (2010). $I(Common\ Education)$, $I(Common\ Gender)$, and $I(Common\ Race)$ are indicator variables equal to one if at least one firm-fund manager pair shares the same undergraduate or MBA institution, gender, or race, respectively. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

5.8 Table 6: impression-management mechanism

Read:

- The timing deterioration is concentrated in *buy* orders, not sells.
- The effect is stronger when CEO delta is higher, meaning managers have stronger incentives to convey favorable information.
- This pattern is hard to reconcile with a generic information-loss story alone and fits the impression-management channel well.

Table 6
Impression management.

	Dependent variable: $DGTW_{RET,j,t+1}$					
	Buy (1)	Sell (2)	Buy (3)	Sell (4)	(5)	(6)
Local					0.010	
					(0.079)	
Lockdown	−1.133*	−1.149*	−0.846	−0.611	−0.988	−1.187*
	(0.579)	(0.675)	(0.521)	(0.556)	(0.620)	(0.649)
ΔExcess w	0.278**	0.051	0.188	0.132	−0.913***	−1.215***
	(0.128)	(0.260)	(0.128)	(0.224)	(0.137)	(0.148)
$DGTW_t$	−0.128***	−0.149***	−0.116***	−0.141***		
	(0.006)	(0.009)	(0.005)	(0.007)		
Local × Lockdown	−0.070	0.166	−0.150	−0.395*	0.022	−0.057
	(0.213)	(0.346)	(0.155)	(0.227)	(0.177)	(0.188)
Local × ΔExcess w	0.217	0.177	0.294	0.060	−0.019	−0.120
	(0.284)	(0.584)	(0.256)	(0.421)	(0.245)	(0.260)
Lockdown × ΔExcess w	0.849***	0.086	0.460**	−0.420	0.369*	0.461*
	(0.215)	(0.377)	(0.230)	(0.330)	(0.213)	(0.236)
Local × CEO Delta					0.00002	
					(0.00005)	
Lockdown × CEO Delta					0.0001	0.0001
					(0.0001)	(0.0001)
ΔExcess w × CEO Delta					0.0003*	0.0003**
					(0.0001)	(0.0001)
Local × Lockdown × ΔExcess w	−0.817*	0.214	−0.876**	0.103	0.134	0.168
	(0.435)	(0.842)	(0.378)	(0.626)	(0.364)	(0.403)
Local × Lockdown × CEO Delta					−0.0001	−0.00003
					(0.0001)	(0.0001)
Local × ΔExcess w × CEO Delta					0.0002	0.0003
					(0.0002)	(0.0002)
Lockdown × ΔExcess w × CEO Delta					−0.0002	−0.0002
					(0.0002)	(0.0002)
Local × Lockdown × ΔExcess w × CEO Delta					−0.001**	−0.001**
					(0.0002)	(0.0002)
Fund × Stock FE	Yes	Yes	Yes	Yes		Yes
Fund FE					Yes	
Stock FE					Yes	
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	552,220	172,970	1,941,365	356,670	1,796,651	1,796,651
R^2	0.294	0.474	0.113	0.275	0.049	0.082

Table 6 Columns (1)–(4) report results from estimating Equation (4) separately for active buy and sell trades relative to the benchmark index fund. Columns (1)–(2) restrict the sample to observations with changes in holdings. Columns (3)–(4) additionally include observations with no change in holdings, capturing benchmark deviation decisions while excluding zero-holding cases. Columns (5)–(6) present results from estimating Equation (4) with the triple-interaction term interacted with *CEO Delta* in 2019 from Core and Guay (2002).

Table 7
Trade activity and local preference.

Panel A. Trading activity

	Dependent variable: Active share	Turnover		ln(Trade size)		Num initiation	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Local			−0.033***		0.011		−0.003***
			(0.001)		(0.032)		(0.0003)
TNA	−0.000***	−0.000***	−0.000**	−0.000***	−0.000***	−0.000***	−0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Flow	0.005	0.001	0.001	0.050	0.054	0.0003	−0.0001
	(0.035)	(0.001)	(0.0005)	(0.039)	(0.041)	(0.0003)	(0.0003)
Return	0.006***	−0.0003***	0.0002*	−0.055***	0.003	−0.001***	−0.00004
	(0.002)	(0.0001)	(0.0001)	(0.003)	(0.012)	(0.00004)	(0.0001)
Lockdown	−0.989***	0.007***	0.001	0.764***	0.169	0.009***	0.003
	(0.125)	(0.001)	(0.003)	(0.042)	(0.270)	(0.0003)	(0.003)
Lockdown × Local			−0.005***		0.002		−0.001
			(0.001)		(0.045)		(0.001)
Fund FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time FE			Yes		Yes		Yes
Observations	23,842	16,911	27,363	22,873	43,272	23,842	45,977
R^2	0.969	0.377	0.379	0.375	0.408	0.111	0.131

Panel B. Local preference

	Dependent variable: Local overweight			
	(1)	(2)	(3)	(4)
Lockdown	0.793***	0.894***		
	(0.161)	(0.163)		
Footprint Drop			0.829***	0.946***
			(0.162)	(0.165)
TNA		−0.001***		−0.001***
		(0.0003)		(0.0003)
Flow		0.004		0.006
		(0.023)		(0.023)
Return		0.001		−0.002
		(0.002)		(0.002)
Lockdown × T2	−0.587***	−0.701***		
	(0.206)	(0.211)		
Lockdown × T3	−0.919***	−1.001***		
	(0.244)	(0.248)		
Footprint Drop × T2			−0.625***	−0.736***
			(0.207)	(0.212)
Footprint Drop × T3			−0.941***	−1.026***
			(0.243)	(0.247)
Fund FE	Yes	Yes	Yes	Yes
Observations	22,343	21,082	22,343	21,082
R^2	0.882	0.884	0.882	0.884

Table 7 Panel A reports regression results comparing portfolio-level trading measures before and after the lockdowns. *Active share* is from Cremers and Petajisto (2009). *Turnover* is the monthly average turnover ratio, calculated as the minimum of aggregate purchases and sales divided by monthly fund assets. *Trade size* is the average dollar value of trades at the portfolio level, and *Num initiation* is the number of initiated positions relative to the number of holdings. Panel B reports regression results comparing local overweighting before and after the lockdowns at the fund level. The dependent variable, *Local overweight* is from Coval and Moskowitz (2001). Funds are grouped into terciles based on their median local bias in 2019, with the lowest tercile (T1) serving as the baseline. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

5.9 Table 7: trade activity and local preference

Read:

- Lockdowns reduce Active Share modestly, but they do not shut down trading activity.
- Funds continue to trade in terms of turnover, trade size, and position initiation.
- Local overweighting converges across funds: managers with strong pre-pandemic local tilt pull back relatively more.
- So the paper is not about managers becoming passive; it is about the erosion of a specific local-information channel.

5.10 Table 8: within-family information flow

Table 8
Speed of information diffusion within fund families.

Panel A. Within-family SID before and after lockdowns

	Dependent variable:		
	SID	SID_{Within}	SID_{Across}
	(1)	(2)	(3)
Lockdown	−0.007	−0.014	0.004
	(0.011)	(0.017)	(0.012)
Fund family FE	Yes	Yes	Yes
Observations	1341	1051	1146
R^2	0.861	0.791	0.790

Panel B. Heterogeneity in investment timing across funds with different SIDs

	Dependent variable:					
	$DGTW_{RET,j,t+1}$					
	SID		SID_{Within}		SID_{Across}	
	(1)	(2)	(3)	(4)	(5)	(6)
Local × Lockdown × Δw × High SID	−0.247	−0.249	0.022	0.020	0.158	0.134
	(0.431)	(0.429)	(0.441)	(0.435)	(0.474)	(0.470)
Fund FE	Yes	Yes	Yes	Yes	Yes	Yes
Stock FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm Controls		Yes		Yes		Yes
Observations	2,122,962	2,110,487	1,951,174	1,939,860	2,044,638	2,032,442
R^2	0.054	0.065	0.054	0.065	0.054	0.065

Panel C. Investment timing in stocks local to colleagues

	Dependent variable:			
	$DGTW_{RET,j,t+1}$			
	(1)	(2)	(3)	(4)
$Local_{Colleague}$	0.063	0.095**	0.063	0.095**
	(0.041)	(0.044)	(0.041)	(0.044)
Lockdown	−1.118**	−0.936*	−1.132**	−0.937*
	(0.528)	(0.510)	(0.529)	(0.510)
Δw	−1.125***	−0.149		
	(0.134)	(0.114)		
ΔExcess w			−0.966***	−0.137
			(0.122)	(0.105)
$Local_{Colleague}$ × Lockdown	−0.160*	−0.235**	−0.160*	−0.235**
	(0.094)	(0.101)	(0.094)	(0.101)
$Local_{Colleague}$ × Δw	0.464***	0.428***		
	(0.139)	(0.136)		
Lockdown × Δw	0.275	0.297		
	(0.201)	(0.194)		
$Local_{Colleague}$ × Lockdown × Δw	−0.536***	−0.551***		
	(0.198)	(0.193)		
$Local_{Colleague}$ × ΔExcess w			0.401***	0.350***
			(0.129)	(0.125)
Lockdown × ΔExcess w			0.187	0.210
			(0.190)	(0.184)
$Local_{Colleague}$ × Lockdown × ΔExcess w			−0.405**	−0.400**
			(0.193)	(0.187)
Fund FE	Yes	Yes	Yes	Yes
Stock FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Firm Controls		Yes		Yes
Observations	2,333,676	2,320,003	2,333,676	2,320,003
R^2	0.054	0.065	0.054	0.065

(continued on next page)

Table 8 (continued).

Table 8 reports results on within-fund-family information flow, measured using the Speed of Information Diffusion (SID) from Cici et al. (2016). Panel A presents results comparing SID before and after the lockdowns, defining the post-lockdown period as 2020 Q2 onward. Standard errors are clustered at the family level. Panel B reports results from estimating Equation (4), interacting the triple-interaction term with $High\ SID$, an indicator equal to one for funds in families with above-median SID in 2019. Standard errors are clustered at the fund and stock levels. Panel C presents results from estimating Equation (4) using $Local_{Colleague}$, an indicator equal to one if a stock is located within 100 miles of at least one affiliated fund but not the focal fund. Standard errors are clustered at the fund and stock levels. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Read:

- SID does not fall significantly after the lockdowns, so internal family communication is not the main story.
- Pre-lockdown SID does not explain the deterioration in timing.
- Stocks local to colleagues do show some negative effect, but it is much smaller than the main local effect.
- This makes within-family information flow a secondary channel at most.

5.11 Table 9: public information seeking

Read:

- Reliance on Public Information changes only modestly after the lockdowns.
- Funds that were initially less reliant on public information increase that reliance somewhat, but the shift is too small to offset the loss of F2F interaction.
- So the main result cannot be explained as managers fully substituting into public information.

Table 9
Public information seeking.

	Dependent variable:			
	RPI_1 (1)	$RPI_{1,dollar}$ (2)	RPI_4 (3)	$RPI_{4,dollar}$ (4)
Lockdown	−0.010***	−0.010***	−0.002	−0.002
	(0.002)	(0.002)	(0.001)	(0.001)
Low RPI	−0.025***		−0.009***	
	(0.001)		(0.001)	
Lockdown × Low RPI	0.018***		0.005***	
	(0.004)		(0.001)	
Low RPI_{dollar}		−0.025***		−0.009***
		(0.001)		(0.001)
Lockdown × Low RPI_{dollar}		0.018***		0.005***
		(0.004)		(0.001)
Fund style FE	Yes	Yes	Yes	Yes
Observations	6636	6636	5805	5805
R^2	0.039	0.039	0.006	0.006

Table 9 reports results for fund managers' Reliance on Public Information (RPI) before and after lockdowns, calculated following Kacperczyk and Seru (2007). RPI is constructed using the number of shares, while RPI_{dollar} is constructed using the dollar value of holdings. Columns (1)–(2) use one lag of analyst recommendations, and Columns (3)–(4) use up to four lags. *Low RPI* denotes the below-median RPI group before the lockdowns. The post-lockdown period is defined as 2020 Q2 onward. Standard errors are clustered at the fund style level. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

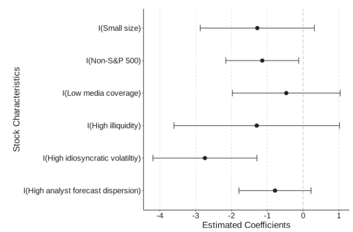


Fig. 6. Stock informational environment.
Fig. 6 presents regression results examining cross-sectional heterogeneity in local investment timing by stock information environment. It shows results from estimating Equation (4), where the triple-interaction term is further interacted with a high-sensitivity dummy that splits sample stocks into two groups based on 2019 median values of: (1) total assets, (2) media mentions, (3) Amihud illiquidity, (4) idiosyncratic volatility, (5) analyst forecast dispersion, and (6) S&P 500 inclusion. The figure plots point estimates and 90% confidence intervals for the main interaction terms, estimated separately in each regression.

5.12 Figure 6: heterogeneity by informational environment

Read:

- The local timing deterioration is more negative for stocks in less transparent environments.
- The strongest significant heterogeneity appears for non-S&P 500 firms and firms with high idiosyncratic volatility.
- All estimated coefficients are negative across the informational-sensitivity proxies, which is exactly what the soft-information view predicts.

5.13 Table 10: regional social characteristics

Read:

- The local-performance decline is stronger in counties with richer social infrastructure and higher job density, where F2F interaction was likely especially important before the pandemic.
- The effect is weaker in areas with rare mask use, interpreted as places where in-person contact persisted more despite the pandemic.
- These regional patterns reinforce the interpretation that the core shock is specifically about disrupted F2F contact.

6 Short summary

- **Conclusion.** COVID lockdowns reduced mutual fund managers' local informational advantage, and the strongest evidence points to worse local trade timing rather than just worse local

Table 10
Regional social characteristics.

	Dependent variable:					
	Benchmark-adjusted return			DGTW-adjusted return		
	(1)	(2)	(3)	(4)	(5)	(6)
Local	-0.369^{***}	-0.328^{***}	-0.097^{**}	-0.289^{***}	-0.271^{***}	0.049
	(0.058)	(0.063)	(0.045)	(0.054)	(0.059)	(0.040)
Footprint Drop	0.618^{***}	0.486^{**}	0.701^{***}	0.811^{***}	0.752^{***}	0.929^{***}
	(0.189)	(0.191)	(0.184)	(0.167)	(0.170)	(0.164)
Local × Footprint Drop	−0.162	−0.151	-0.616^{***}	−0.043	0.030	-0.335^{***}
	(0.102)	(0.106)	(0.074)	(0.092)	(0.098)	(0.068)
Local × High Social Index	0.151^{**}			0.225^{***}		
	(0.074)			(0.068)		
Footprint Drop × High Social Index	−0.0004			0.081		
	(0.067)			(0.057)		
Local × Footprint Drop × High Social Index	-0.426^{***}			-0.259^{**}		
	(0.128)			(0.117)		
Local × High Job Density		0.069			0.180^{**}	
		(0.075)			(0.070)	
Footprint Drop × High Job Density		0.246^{***}			0.192^{***}	
		(0.067)			(0.057)	
Local × Footprint Drop × High Job Density		-0.429^{***}			-0.387^{***}	
		(0.131)			(0.120)	
Local × Rare Mask Use			-0.425^{***}			-0.491^{***}
			(0.075)			(0.069)
Footprint Drop × Rare Mask Use			-0.170^{**}			-0.157^{***}
			(0.067)			(0.057)
Local × Footprint Drop × Rare Mask Use			0.523^{***}			0.342^{***}
			(0.133)			(0.122)
Fund FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	47,606	47,606	47,606	47,606	47,606	47,606
R^2	0.082	0.082	0.083	0.087	0.087	0.088

Table 10 reports heterogeneous fund-level performance results across regions with varying social characteristics. Equation (2) is re-estimated by interacting the DiD term with the following social characteristics: (1) *High Social Index* denotes the above-median group based on the county-level social index (*ASSN* 2014), which counts local organizations as a proxy for social infrastructure. (2) *High Job Density* denotes the above-median group based on destination accessibility (*D5AR*), which measures the number of jobs reachable within 45 min. (3) *Rare Mask Use* denotes the above-median group based on the share of survey respondents reporting rarely wearing masks in public. Higher values indicate lower compliance with social distancing guidelines. Standard errors are clustered at the fund level. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

fundamentals.

- **Mechanism insight.** F2F interaction matters because it helps transmit soft information through trust-building and impression management.
- **Empirical lesson.** Even in modern capital markets, some information channels are not well replicated by remote communication or public disclosure.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes a causal role for F2F interaction in generating local informational advantage for mutual fund managers.
- The key empirical contribution is not just showing that local investors lost an edge during COVID, but showing *why*: their timing on the same stocks worsened when in-person interaction became harder.
- The paper also shows that the mechanism is consistent with two classic soft-information channels—trust-building and impression management—rather than with generic trading slowdowns or simple shifts toward public information.

7.2 Economic intuition

Local informational advantage is not just about being geographically close. It is about what proximity *allows investors to do*. Before the pandemic, being near firms made it easier to meet managers, observe tone and confidence, build trust, and pick up nuanced qualitative information. Once that channel was interrupted, local investors were still geographically close in a literal sense, but the effective informational advantage from proximity weakened. That is why the paper interprets the lockdown shock as a test of the *mode of communication*, not just the location of the investor.

The mechanism results sharpen this intuition. Low-trust relationships suffer more because they depend more on in-person trust formation. Buy decisions suffer more because favorable soft information is more actionable there, especially when managers cannot short. And the effects are strongest in opaque informational environments, where soft information is relatively more valuable.

7.3 Takeaway

This paper is best understood as a clean design for separating *locality* from *face-to-face interaction*. Its main message is that proximity matters in finance partly because it enables in-person contact, and that in-person contact still carries unique informational value. For research on mutual funds, local bias, soft information, corporate access, remote work, or the informational role of social interaction in markets, this paper is now a benchmark reference.